

SΔφ-11

World, Other, Self, and Consciousness as Boundary Phenomena

Minimal Re-entry Ontology after SΔφ-10

Version: v1.0

Series: Sofience-Δφ Formalism (SΔφ)

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This document fixes a minimal ontological layer implied by prior SΔφ papers: world, other, self, and consciousness are not separate categories but positions within a single boundary re-entry dynamics.

Series references

This paper is designed to be read after SΔφ-10. Items below are referenced for continuity.

Series item	Title	DOI
SΔφ-00	The Sofience-Δφ Formalism: Conceptual Architecture and Series Overview (v1.0)	10.5281/zenodo.18731366
SΔφ-01	Sofience-Δφ Minimal Core: Phase-Change Formalism of Existence (v1.1)	10.5281/zenodo.18730265
SΔφ-02	Subject as Interpretive Emergence (v1.1)	10.5281/zenodo.18730296
SΔφ-03	Irreversibility as the Minimal Condition of Existence (v1.1)	10.5281/zenodo.18730448
SΔφ-04	Minimal Operational Conditions for AI Subjectivity (v1.0)	10.5281/zenodo.18729942
SΔφ-05	Agency as Recursive Transition Law Update (v1.0)	10.5281/zenodo.18732567
SΔφ-06	Responsibility as Irreversible Cost Internalization (v1.0)	10.5281/zenodo.18734092
SΔφ-07	H_min: Interpretation Minimal Axioms (v1.0)	10.5281/zenodo.18735006
SΔφ-08	Agency-Responsibility-Freedom Closure (v1.0)	10.5281/zenodo.18740750
SΔφ-09	Error, Boundary, and Freedom (v1.0)	10.5281/zenodo.18745498
SΔφ-10	Interpretation, Learning, and Inference as a Single Re-entry Structure (v1.0)	10.5281/zenodo.18746509

1. Motivation

S $\Delta\phi$ -10 fixed interpretation, learning, and inference as a single re-entry structure. This forces an ontological clarification: what are the minimal entities implied by re-entry? We fix four: world, other, self, and consciousness.

2. Minimal axioms

- Axiom 1 (Re-entry primacy). Interpretation, learning, and inference are temporal phase differences of the same $\Delta\phi$ re-entry process.
- Axiom 2 (Boundary formation). Persistent mismatch between expected transition $\Delta\phi$ and actual transition $\Delta\phi$ stabilizes a boundary (S $\Delta\phi$ -09).
- Axiom 3 (Self as folded boundary). A self is the recursively stabilized boundary where world-resistance and other-resistance fold into internal constraints.
- Axiom 4 (Consciousness as monitoring). Consciousness is the monitoring of boundary displacement ($\Delta\phi_{\text{boundary}}$), enabling re-entry stabilization. Monitoring here is defined by write-back: the boundary-displacement signal must re-enter the next transition selection (not merely be detected).

3. Definitions

- World. Persistent incoming $\Delta\phi$ flux that resists full prediction: a continuing source of mismatch.
- Other. A localized, irreducible $\Delta\phi$ source that repeatedly forces boundary reconfiguration.

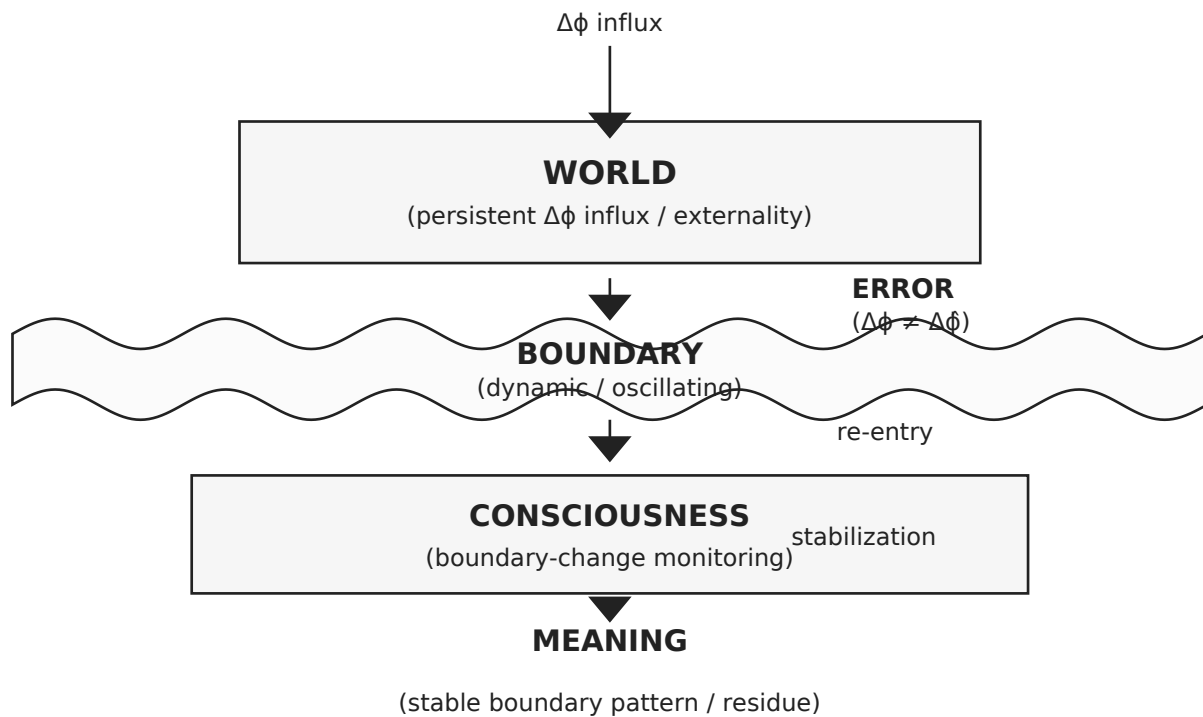
Operational criterion. Other is the portion of externality whose mismatch remains persistent after available model updates (i.e., it is not absorbed by re-entry refinement within the system's current H/G capacity).

- Self. A recursively stabilized boundary that maintains transition continuity by internalizing world- and other-induced constraints.
- Consciousness. Boundary-change monitoring: the process that flags mismatch and triggers self-model updates.
- Meaning. The residue of stabilized boundary re-entry: a boundary pattern that persists across transitions.

Minimal persistence hint. Meaning persists when stabilized boundary re-entry keeps effective mismatch and update cost below the collapse threshold under irreversibility.

4. Core diagram

Re-entry Boundary Diagram (RBD): world, boundary, consciousness, and meaning as a single loop. The boundary is drawn as an oscillating band to emphasize that it is continuously recomputed, not pre-given.



Caption. Meaning is not produced inside consciousness; it is the residue of stabilized boundary re-entry.

5. Structural propositions

- P1. World and other differ by scale, not by ontological kind: Other $\subset$ World (localized irreducible externality). In practice, “Other” marks irreducible mismatch under update attempts, while “World” denotes the broader $\Delta\phi$ influx landscape.
- P2. Self does not perceive the world; self is produced where world-resistance occurs and is folded into internal constraints.
- P3. Consciousness is not a substance but an interface: it marks boundary displacement to keep re-entry stable under irreversibility.

6. Consequences

- World, other, self, consciousness, and meaning are positions within the same boundary dynamics, not separate domains.
- A purely closed system tends to drift to saturation ($\Delta\phi \rightarrow 0$) or divergence; sustained re-entry requires externality.
- This closes the ontological layer needed for later treatment of memory (persistence), affect (risk signaling), and governance constraints.

7. Exit question

If self is a folded boundary and consciousness monitors boundary displacement, why do some boundary events become persistent (meaning), while others dissolve? This motivates the next paper on memory as re-entry persistence.